

Logistic Regression Analysis

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Classification – Logistic Regression I

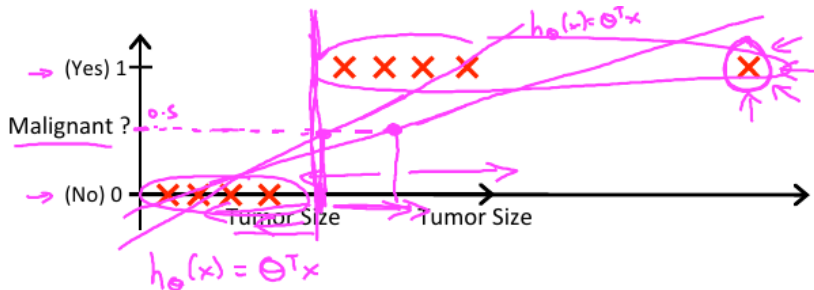
- Email: Spam / Not Spam?
- Online Transactions: Fraudulent (Yes/No)?
- Tumor: Malignant / Benign?

$$y \in \{0, 1\}$$

0: “negative class” (e.g. benign tumor)

1: “positive class” (e.g. malignant tumor)

Classification – Logistic Regression II



→ Threshold classifier output $h_{\theta}(x)$ at 0.5:

→ If $h_{\theta}(x) \geq 0.5$, predict "y = 1"

If $h_{\theta}(x) < 0.5$, predict "y = 0"

Classification – Logistic Regression III

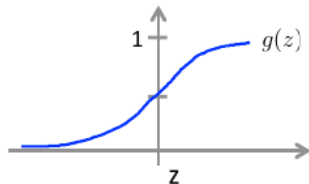
- Target output for classification: $y = 0$ or $y = 1$
 - Regression Analysis (Classification): $h_{\theta}(x)$ can be >1 or <0
 - Logistic Regression (Classification): $0 \leq h_{\theta} \leq 1$

Model (Hypothesis) Representation

Logistic regression

$$h_{\theta}(x) = g(\theta^T x)$$

$$g(z) = \frac{1}{1+e^{-z}}$$



Suppose predict “ $y = 1$ ” if $h_{\theta}(x) \geq 0.5$

$$\theta^T x \geq 0$$

$$g(z) \geq 0.5 \\ \text{when } z \geq 0$$

$$h_{\theta}(x) = g(\theta^T x)$$

predict “ $y = 0$ ” if $h_{\theta}(x) < 0.5$

$$\theta^T x < 0$$

$$g(z) < 0.5 \\ \text{when } z < 0$$

- g is Sigmoid (Logistic) function.
- It clamps the output of the linear equation between 0 and 1.
- The output of g indicates the probability of the presented inputs to be $y = 1$ (i.e. being malignant tumor).

Cost Function I

Training
set:

$$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$$

m examples

$$x \in \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix}$$

$\mathbb{R}^{n+1}$

$$x_0 = 1, y \in \{0, 1\}$$

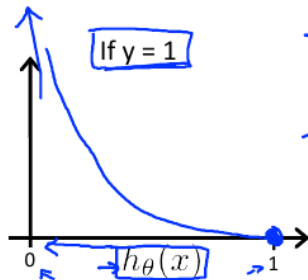
$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

How to choose parameters θ ?

Cost Function II

Logistic regression cost function

$$\text{Cost}(\underline{h_{\theta}(x)}, y) = \begin{cases} \boxed{-\log(h_{\theta}(x))} & \text{if } y = 1 \\ \underline{-\log(1 - h_{\theta}(x))} & \text{if } y = 0 \end{cases}$$



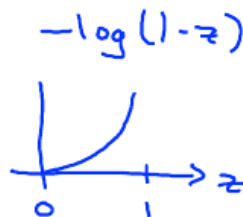
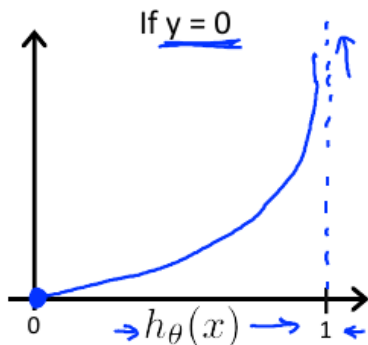
→ Cost = 0 if $y = 1, h_{\theta}(x) = 1$

But as $\frac{h_{\theta}(x) \rightarrow 0}{\text{Cost} \rightarrow \infty}$

→ Captures intuition that if $h_{\theta}(x) = 0$, (predict $\underline{P(y = 1|x; \theta) = 0}$), but $\underline{y = 1}$, we'll penalize learning algorithm by a very large cost.

Logistic regression cost function

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



Cost Function IV

Logistic regression cost function

$$\begin{aligned} J(\theta) &= \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) \\ &= \frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right] \end{aligned}$$

To fit parameters θ :

$$\min_{\theta} J(\theta) \quad \text{Get } \underline{\theta}$$

To make a prediction given new $\underline{x}$:

$$\text{Output } \underline{h_{\theta}(x)} = \frac{1}{1 + e^{-\theta^T x}}$$

$$\underline{p(y=1 | x; \theta)}$$

Gradient Descent Algorithm

Gradient Descent

$$\rightarrow J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want $\min_{\theta} J(\theta)$:

Repeat {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

}

(simultaneously update all θ_j)

$$\frac{\partial}{\partial \theta_j} J(\theta) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

Multi-Class Classification I

Email foldering/tagging: Work, Friends, Family, Hobby

$y=1$ $y=2$ $y=3$ $y=4$

Medical diagrams: Not ill, Cold, Flu

$y=1$ 2 3

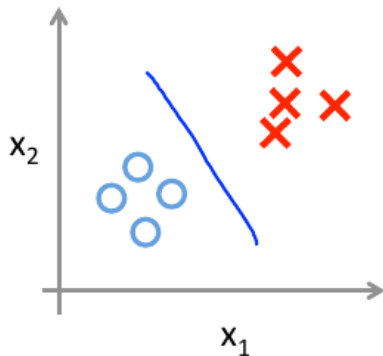
Weather: Sunny, Cloudy, Rain, Snow

$y=1$ 2 3 4 ←

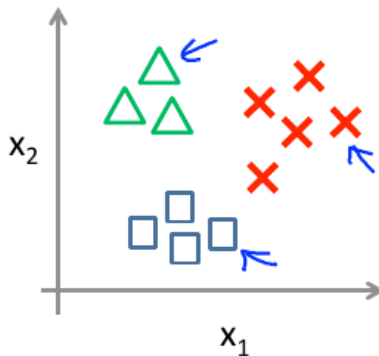


Multi-Class Classification II

Binary classification:

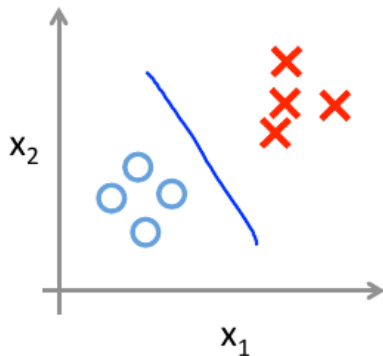


Multi-class classification:

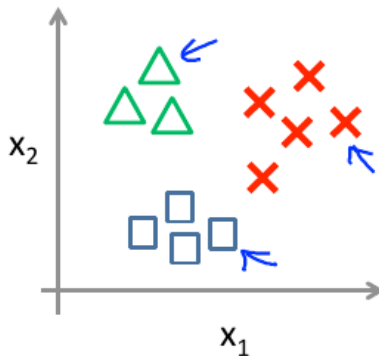


Multi-Class Classification III

Binary classification:



Multi-class classification:



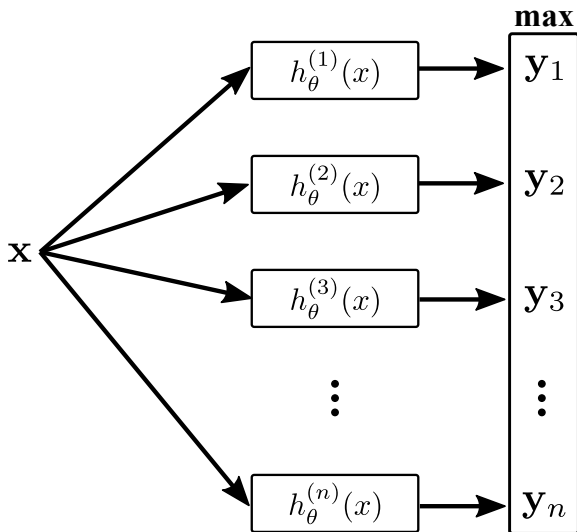
Multi-Class Classification IV

- Train a logistic regression $h_{\theta}^{(i)}(x)$ for each class i to predict the probability that $y = i$.
- On a new input x to make a prediction, pick the class i that maximises
$$\arg \max_i h_{\theta}^{(i)}(x)$$

Multi-Class Classification V

INPUT		OUTPUT		
		Object	Class index	One-hot vector
image 1	$\mathbf{x}^{(1)}$	car	1	$y_1, y_2, y_3 \dots, y_n$ 1 0 0 ... 0 0 0
image 2	$\mathbf{x}^{(2)}$	bus	2	0 1 0 ... 0 0 0
image 3	$\mathbf{x}^{(3)}$	bird	3	0 0 1 ... 0 0 0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
image m	$\mathbf{x}^{(m)}$	dog	n	0 0 0 ... 0 0 1

Multi-Class Classification VI



Multi-Label Classification I

	Multi-Class	Multi-Label
C = 3		
	Samples   	Samples   
		
		
	Labels (t) [0 0 1] [1 0 0] [0 1 0]	Labels (t) [1 0 1] [0 1 0] [1 1 1]

